Dual-threshold time-walk calibration for CMS BTL prototypes

CMS

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Discussion draft; not approved for public release.

Abstract

A charge-dependent time-walk model is studied for CMS barrel timing layer (BTL) prototype channels using two discriminator thresholds. Their current separation divided by the measured crossing-time separation provides a finite-interval pulse-slope proxy. A linear profile fit describes this proxy as a function of the measured charge, and an inverse-slope function models the timing correction with one amplitude parameter per channel.

Slope measurements from run 4076 are compared with independent microchannel-plate (MCP) reference calibrations from run 4399. The two runs contain different events. A free constant offset and the correction amplitude are fitted only where both measurements have populated charge support. The resulting agreement is a calibration-shape comparison, not an independent measurement of timing resolution.

The draft also examines channel-to-channel variations and an exploratory prediction of the correction amplitude from internal pulse information. Such a prediction could enable operation without an event-by-event external time reference, but its transferability and full uncertainty remain to be established.

Two complementary runs, not the same events

Sample	Available information	Role in this study
Run 4076	Valid dual-threshold timing and charge	Measure the pulse-slope proxy $S(Q)$
Run 4399	MCP-associated reference timing and charge	Measure the empirical walk $W_{\mathrm{MCP}}(Q)$

The runs contain different particles and are never matched event by event. The comparison transfers a channel-response model between runs.

The upstream and downstream modules each contain bars read out at two ends. Detector/readout context is given in Refs. [1, 2]. The channel/bar mapping determines the coincidence pairs; the same channel number is compared across the two runs.

The comparison assumes the same nominal detector setup in both runs, with MCP reference timing in run 4399 and dual-threshold timing in run 4076. Cross-run response differences are not included in the quoted uncertainties.

From two crossing times to a walk model

For thresholds $I_1 < I_2$ and their valid crossing times t_1, t_2, define

$$S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}], \qquad \overline{S}(Q) \simeq c_0 + c_1 Q.$$

S is a finite-interval rising-edge slope proxy. It is not the current difference ΔI itself and is not necessarily the instantaneous derivative at the first threshold.

$$W_Q(Q) = \frac{A}{c_0 + c_1 Q}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W_Q(Q).$$

The amplitude A (AWC) has units of current and is constant for a channel and operating configuration. The correction varies event by event through Q.

Timing reconstruction and threshold scale

The run-4076 sample is file `23_e.root`. For coarse codes C_1, C_2,

$$d_c = (C_2 - C_1) \bmod 1024, \qquad t_2 = t_1 + (6250\,\mathrm{ps})d_c - f_1 + f_2.$$

The coarse codes use signed 64-bit arithmetic before subtraction; the coarse time term uses floating point, avoiding the old 16-bit multiplication overflow.

The fine-time lookup stores the mean of t_1 mod $6250\,\mathrm{ps}$ for each (channel, TAC, fine-code) key, using all input hits. The same lookup supplies f_1 and f_2. No selected hit in the 22 audited channels has a missing entry.

All selected hits have threshold register `step2=113621`, decoded as DAC codes 10 and 35. The implemented scale gives

$$\Delta I = (35 - 10) \times 0.3125\,\mu\mathrm{A} = 7.8125\,\mu\mathrm{A}, \qquad \frac{S}{\mu\mathrm{A/ns}} = 1000\,\frac{\Delta I/\mu\mathrm{A}}{(t_2 - t_1)/\mathrm{ps}}.$$

Implementation: `Q_Walk/qwalk_core.py`. This checks the implemented conversion, not an independent hardware-threshold calibration.

Selection and profile construction

- Require one hit at each end of the target bar and no other hit in that module. The other module is unconstrained; channel 192 is ignored.
- Require finite Q, S, $\Delta I > 0$, $t_2 - t_1 > 0$ and $d_c = 0$. This selects the same-coarse-bin prompt population.
- Require $Q \geq Q_{\mathrm{left}}$, the left half-height boundary of the smoothed charge histogram. The Landau–Gaussian fit does not set this cut. There is no additional upper-charge cut or 100 ps minimum-rise cut.
- Form ten equal-width bins over the selected charge range. Plot mean S and hit SD; fit with SEM weights. Bins with undefined or negligible SEM are omitted.
- Bootstrap selected hits with fixed bin edges and refit 1000 times. Quote the 16th–84th percentiles; Ref. [4] gives the general principle.

All selected hits are used; there is no train/test split. Implementation: `Q_Walk/qwalk_two_hit_pair_profile.py`, with the newer optional minimum-rise cut disabled.

Reproducing the frozen run-4076 profiles

Replaying `23_e.root` reproduces the selected counts, ten-bin profiles and linear-fit coefficients for all 22 channels. Saved bootstrap coefficients reproduce their reported percentile intervals.

Channel	Target-pair hits	Prompt hits	Peak-right hits
137	25979	20789	19126

For channel 137, the histogram-derived boundary is $Q_{\mathrm{left}} = 184.9163$ ADC. Charge-bin centers are the mean measured Q, not the geometric bin centers.

The frozen selection retains rises below 100 ps in channels 130, 139, 151, 159, 236 and 238. These events were not removed retrospectively. Their effect belongs to the selection-sensitivity and channel-quality studies.

Channel 135 has one singleton bin: its SEM is undefined and only nine bins enter that channel's linear fit. No fit or plotted data changed in this audit.

Audit, input/code hashes and per-channel counts: `DPS_NOTES/review_2026-10-22_n2_2/`.

Reference version shared with Note 1

The run-4399 MCP functions were refitted on 22 October 2026 after correcting segment detection across missing trigger timestamps and excluding invalid clock groups. Note 1 documents the timing validation and remaining limitations.

For each topology, file and segment, the scaled MAD of the channel-192 versus scope-trigger line residuals must not exceed 1 ns before LOWESS [3]. The selection uses clock information, not SiPM timing residuals. Suspect event associations are excluded; no MCP rows are shifted.

All AWC-and-offset fits in this note use these refreshed references. The run-4076 slope fits, bootstrap replicas and measured-support matching definition are unchanged. The charge spectra use the updated reference sample.

AWC intervals still hold the reference fixed. They do not account for reference-fit uncertainty, clock-selection uncertainty or cross-run transfer.

Numerical version: `Q_Walk/results/2026-10-22_clock_quality`.

Sensitivity to the refreshed reference

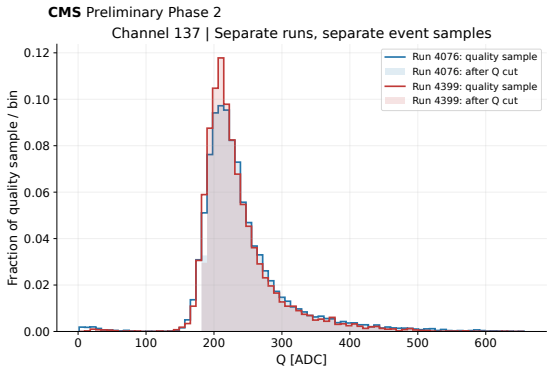
The run-4076 input is fixed; these changes arise from the reference rerun, including its automatically determined charge selection and measured support.

Channel	Previous AWC [μA]	Refreshed AWC [μA]
137	18.513	18.332
136	27.899	-4.240
151	7.649	9.354

Channel 136 has only 210 selected reference hits. Its sign change, and the 22% shift for channel 151, expose sensitivity not covered by slope-only errors. They are retained in the exploratory summary, not accepted as robust calibrations.

Sparse clock groups retain the existing line-only fallback when fewer than ten alignment inliers are available. A small residual scale in such a group does not independently validate its event association. Reference stability and adequate clock statistics remain prerequisites for a future quality-selected performance sample; this note reports exploratory fits.

Charge spectra define the available comparison sample

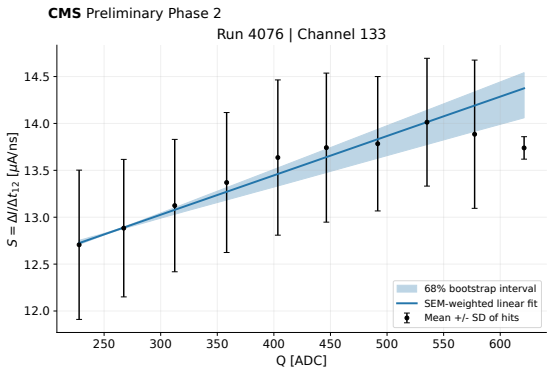


Channel 137 charge spectra are shown for the two independent runs. Lines show quality-selected hits; shaded spectra retain the additional charge-peak selection. Each run is normalized to its own quality-sample count.

The charge proxy is in ADC units, not MeV. Different populations and acceptances motivate restricting the curve alignment to jointly supported charge values.

Recovered histogram counts, not a new peak fit. Run-4076 profile selection starts at about 184.9 ADC for this channel.

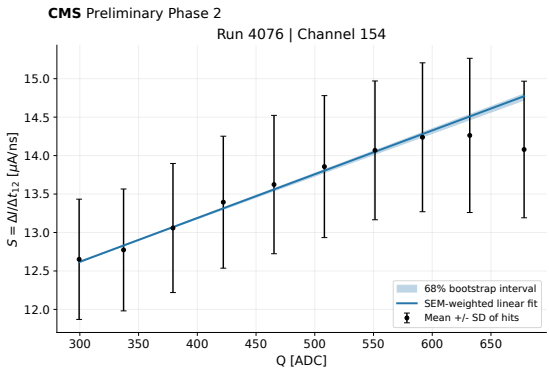
Channel 133: fitting the slope profile



Run 4076 channel-133 hits are grouped into ten charge bins. Points show mean S; bars show the SD of individual selected hits. The line is the saved SEM-weighted fit $c_0 + c_1 Q$.
The shaded interval is the central 68% bootstrap interval of the fitted mean curve. It is shown at its true size, without visual enlargement.

Extended-profile snapshot. SD describes hit spread; SEM and the fit band describe different statistical quantities.

Channel 154: the opposite end of bar 7

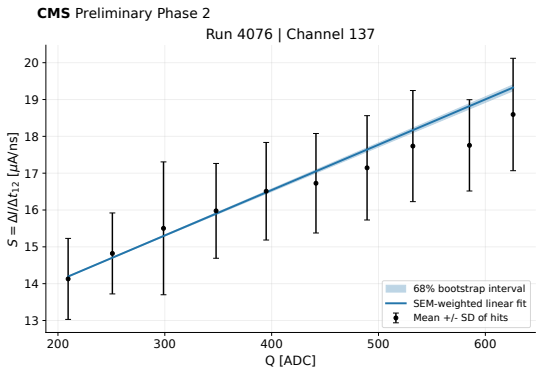


The same selection and profile procedure is applied to channel 154. The mean slope increases with charge, but the response is calibrated independently at each bar end. Bars show hit SD. The line uses SEM weights and the band propagates the saved bootstrap coefficient replicas, including their correlation.

Review N2-3: Linear-model adequacy remains open. The separate sensitivity audit is retained outside this note; this version keeps the original comparison.

Run 4076, extended-profile snapshot. No model refit is performed for the slide rendering.

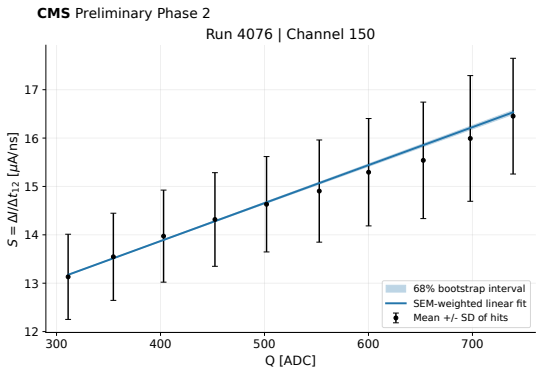
Channel 137: pulse slope and charge



The selected run-4076 sample contains 19 126 channel-137 hits. Ten profile points constrain a linear pulse-slope model over the observed charge range. The SD bars show event-to-event spread, while SEM weights determine the fit. The narrow shaded band is the conditional 68% bootstrap interval of the mean prediction, not a single-hit uncertainty.

Selected count and fit are read from the saved extended-profile metadata. No uncertainty band is magnified.

Channel 150: the opposite end of bar 8



Channel 150 is treated with the same profiling procedure. The charge-dependent slope provides the denominator of the proposed walk function; a separate amplitude A determines the correction strength. The line and shaded interval describe the fitted mean. Points and SD bars retain information on the spread within the selected charge bins.

Run 4076, extended-profile snapshot. Constant AWC does not mean a constant timing correction.

Determine AWC using only jointly measured support

The run-4399 reference is a fitted quadratic in inverse charge,

$$W_{\mathrm{MCP}}(Q) = a + b/Q + c/Q^2.$$

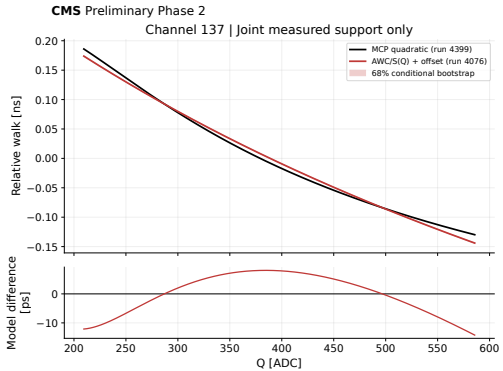
Let Ω be the intersection of the two outer profile-mean ranges, restricted to charge bins populated in both runs. Fit

$$(A, B) = \arg\min_{A,B} \sum_{Q_k \in \Omega} \left[\frac{A}{c_0 + c_1 Q_k} + B - W_{\mathrm{MCP}}(Q_k) \right]^2 .$$

The saved implementation samples a uniform 3000-point charge grid before removing unsupported intervals. The offset B absorbs the arbitrary time zero.

This is uniform-charge shape matching, not an event-weighted likelihood. It uses neither extrapolated tails nor a single central anchor point. Weighting sensitivity and correlated MCP-fit uncertainty are evaluated in the uncertainty check below. Uniform Q remains the nominal definition, not a uniquely optimal estimator.

Channel 137: inverse-slope and MCP walk shapes

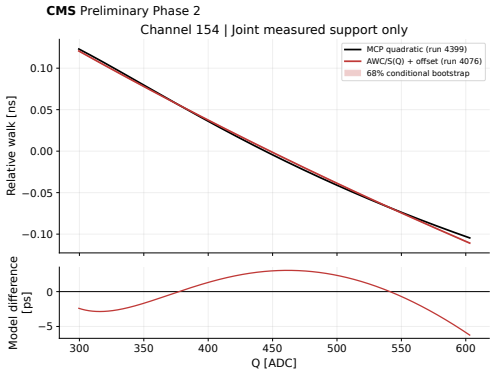


The black curve is the run-4399 MCP quadratic. The red curve is the run-4076 inverse-slope model after fitting AWC and a constant offset. Only jointly measured charge support is shown.

Lower panel: run-4076 model minus MCP model. Its RMS shape mismatch is 6.80 ps on the supported uniform-Q grid, not an event timing resolution. The original band varies the run-4076 slope fit while holding the MCP curve fixed. The later uncertainty table also includes MCP-fit uncertainty.

AWC = 18.332 μA; conditional 16–84% interval [18.120, 18.607] μA. Both curves share one display offset.

Channel 154: a second shape-comparison example

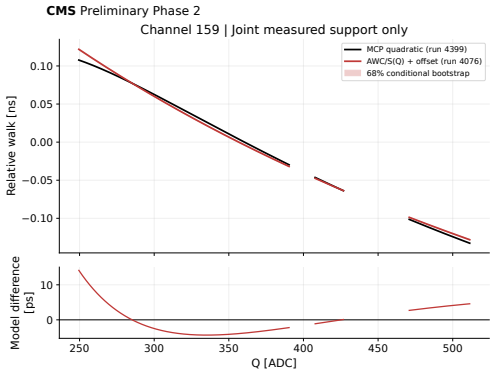


The same overlap-only objective determines AWC and the offset for channel 154. Both plotted curves are calibrated models, not matched events from different runs.

The lower panel directly displays the remaining shape mismatch, with no additional line subtracted. This measures model-shape closure, not corrected-event timing resolution.

Conditional bootstrap band: 4076 slope replicas, refitted AWC and offset, fixed 4399 quadratic.

Channel 159: missing support is not extrapolation

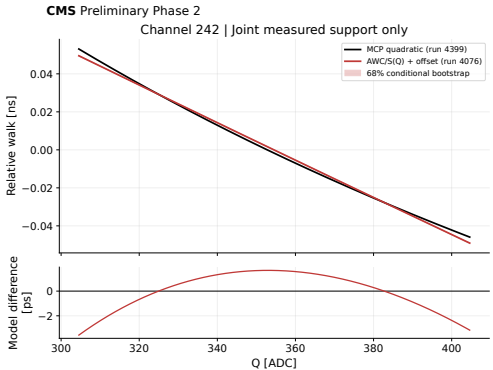


Only intervals with populated profile bins in both runs contribute to the alignment. Breaks in the displayed curves mark excluded charge intervals. They do not represent discontinuities in either analytic model.

The lower panel uses exactly the same supported points. All displayed intervals participate in the common AWC-and-offset fit. It is a model difference, not an event-time distribution.

The stored comparison excludes 703 of the 3000 initial grid points for this channel. No line is drawn through those gaps.

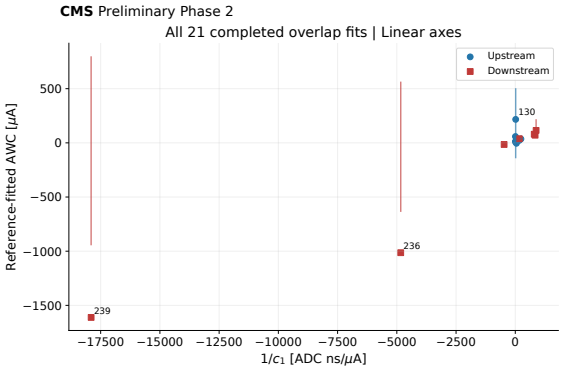
Downstream example: channel 242



The method is also applied to a downstream channel. In run 4399, upstream bar 8 supplies the coincidence pair alongside the downstream target pair and MCP indicator.
In run 4076, the target downstream pair is required, without fixing an upstream pair. The curves use only their common measured support. The same nominal-setup assumption is used for downstream channels.
Selection-dependent response differences remain unquantified.

The lower panel is a model difference, not a timing-resolution measurement. The other module is not used as an internal timing reference to determine AWC here.

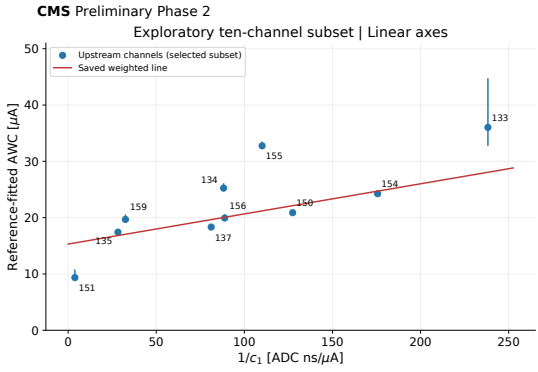
AWC versus inverse slope coefficient: all completed fits



Each point is one channel, not one charge bin. Blue circles are upstream and red squares downstream. The x coordinate is $1/c_1$, the inverse coefficient of the linear $S(Q)$ fit; the y coordinate is reference-fitted AWC. Bars span the 16th–84th bootstrap percentiles. There are 21 completed fits; channel 238 has no measured overlap. All completed fits are retained, including limited-support and poorly constrained cases documented in the channel-status table below.

Linear axes. Large intervals and negative coefficients are retained; this is not a final quality-selected performance sample.

A selected subset does not establish a universal law



The ten-channel upstream subset previously selected for discussion is shown on linear axes. Bars are conditional AWC bootstrap intervals; channel numbers identify the points. The saved weighted straight line has $\chi^2/\mathrm{ndf} = 538.6/8$ under its limited error model. It does not adequately describe the variation. The subset was selected interactively and is illustrative, not quality-selected or an unbiased population-level validation.

The fit treats x as fixed and uses half the 16–84% AWC span as its symmetric uncertainty. Correlations and reference errors are omitted.

Channel coverage and limitations

The explored set contains 14 upstream and 8 downstream channels. All 21 completed alignments are shown; only channel 238 lacks measured overlap. No additional quality cut is applied to this summary.

Channels	Observed limitation and treatment
238	No common measured charge range; no AWC estimate.
235, 244	Overlap spans only 3.10 and 16.99 ADC, respectively; one profile mean from each run lies within each overlap. Fits retained.
130, 236, 239	Slope-only AWC intervals include zero and span hundreds of μA. Fits retained as poorly constrained cases.
136, 245	Negative central AWC values are retained, not accepted as validated positive-amplitude calibrations.

The ten-channel illustration is unchanged: 133, 134, 135, 137, 150, 151, 154, 155, 156 and 159. Its membership is not a derived eligibility rule.

Table entries are diagnostic examples, not additional exclusions. Small curve mismatch over narrow support does not validate a calibration. The full per-channel support and interval table is saved with the figure data.

What the present uncertainties do and do not include

Displayed quantity	Meaning
Profile bars	SD of selected hits in each charge bin; not the uncertainty on the mean.
Fit weights	SEM of each profile mean; not the SD.
Slope band	16–84% range from 1000 bootstrap linear fits with fixed bin edges.
AWC bars / walk band	Propagate the slope replicas through a new AWC-plus-offset alignment to the same fixed reference curve.

These definitions refer to the original displayed bands and bars. The next table additionally propagates the run-4399 quadratic covariance. Cross-run response transfer, clock/fine-time calibration, selection/model choices and cross-channel correlations remain outside both prescriptions. Neither is a total calibration uncertainty or a timing-resolution estimate.

MCP-fit uncertainty and charge weighting

For each slope-bootstrap replica, propagate the correlated MCP coefficient covariance through the AWC-plus-offset fit. The Gaussian mixture gives the 16–84% interval; central AWC values and comparison curves are unchanged.

All table values are in μA. "Both runs" adds MCP-fit statistics, not cross-run or model systematics. The original plots retain slope-only bands.

Channel	AWC	Slope only	Both runs	Max. weight shift
133	36.03	[32.73, 44.75]	[32.48, 45.72]	4.11
154	24.25	[23.92, 24.86]	[22.98, 25.84]	0.85
137	18.33	[18.12, 18.61]	[17.91, 18.80]	2.05
150	20.89	[20.65, 21.15]	[20.41, 21.38]	1.13

Maximum weighting shift: largest $|\Delta\mathrm{AWC}|$ using uniform $1/Q$ or either run's charge population instead of uniform Q. Same overlap and free offset; these shifts are not statistical errors or added in quadrature.

All 21 overlap channels were checked; channel 238 has no overlap. A shared reference-time offset affects only the fitted offset, not AWC.

Review N2-4: Requested audit complete. Accept conditional statistical intervals with separate weighting sensitivity; a total uncertainty budget remains outside this check.

Result and next validation step

- A dual-threshold measurement provides a charge-dependent slope proxy; an inverse-slope function can be aligned to independent MCP walk curves.
- Restricting alignment to jointly measured charge support avoids determining AWC from extrapolation.
- Reference-calibrated AWC varies between channels. A straight line against $1/c_1$ is insufficient for the explored subset.
- The present calibration still uses the MCP to determine AWC. It does not establish reference-free AWC extraction by itself.
- An internal predictor must be tested on held-out bars/configurations with transfer and target uncertainties included before use without MCP.

The following backup explores such a predictor; it is not a final performance claim or a demonstration with cosmic rays.

Backup: an internal predictor for channel AWC

For $W = A/S$ and $S = c_0 + c_1 Q$,

$$-\frac{dW}{d\ln Q} = \frac{A\, c_1 Q}{S(Q)^2}.$$

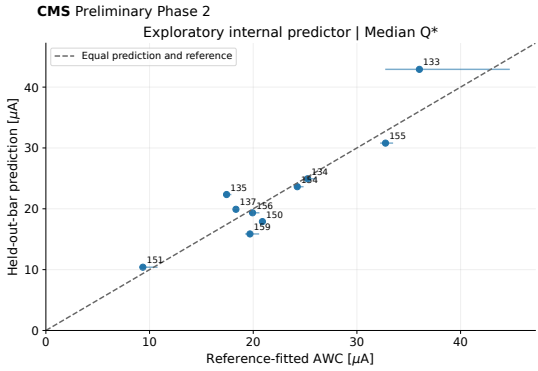
If this local walk derivative is approximately shared between channels at a prescribed typical charge Q_*, a candidate model is

$$A_{\mathrm{pred}} = K\, \frac{S(Q_*)^2}{c_1 Q_*}.$$

Here Q_* is the median charge of the run-4076 quality- and peak-right-selected sample. It is computed once per channel; A_{pred} remains constant when correcting individual events.

The common K is learned from MCP-calibrated training channels. Predicting a new channel uses internal features, but the model development is reference-based. The shared derivative and choice of Q_* remain hypotheses requiring independent validation across operating conditions, without tuning to test scores.

Backup: predict a bar excluded from the model fit



Each point compares reference-fitted AWC with a prediction made after excluding that channel's entire bar from the fit of K. Ten upstream channels span six held-out bar groups.
The RMS prediction error is 3.22 μA; the RMS fractional error is 14.1%.
Horizontal intervals are conditional reference-AWC intervals, not predictive intervals. This small selected sample leaves feature/target correlations and full prediction uncertainties unpropagated. It does not demonstrate reference-free calibration.

Exploratory median-Q_* candidate. The dashed line indicates equality, not a fitted result. No blanket 10% precision claim is made.

References

[1] CMS Collaboration, *A MIP Timing Detector for the CMS Phase-2 Upgrade*, CERN-LHCC-2019-003, CMS-TDR-020 (2019). CERN document record.

[2] E. Albuquerque et al., *TOFHIR2: The readout ASIC of the CMS Barrel MIP Timing Detector* (2024). arXiv:2404.01208.

[3] W. S. Cleveland, *Robust Locally Weighted Regression and Smoothing Scatterplots*, J. Am. Stat. Assoc. **74** (1979) 829–836. doi:10.1080/01621459.1979.10481038.

[4] B. Efron, *Bootstrap Methods: Another Look at the Jackknife*, Ann. Statist. **7** (1979) 1–26. doi:10.1214/aos/1176344552.

These sources provide detector/readout context and methodological background. The run-specific selections, implementations and numerical results are those of this study, not measurements or prescriptions taken from these references.